

Executive Decision Report

Retail Orders — Business Intelligence Capstone

Decision lens: protect realized revenue, target discounts selectively, improve payment conversion, and scale evidence before major pricing changes.

Dataset: 11 unique orders after order-ID deduplication • Net revenue: ■11,240 • Average order: ■1,124

1. Executive summary

What matters now: only 11 unique orders are available, so findings are directional.

- Net revenue: ■**11,240**; median order value: ■1,219.
- Largest category: **Course Access** (■5,546 net revenue).
- Highest average-order segment: **Professional** (■1,575).
- 45.5% of orders are discounted; discount value totals ■2,747.
- Paid-order rate: 63.6%.

2. Methodology

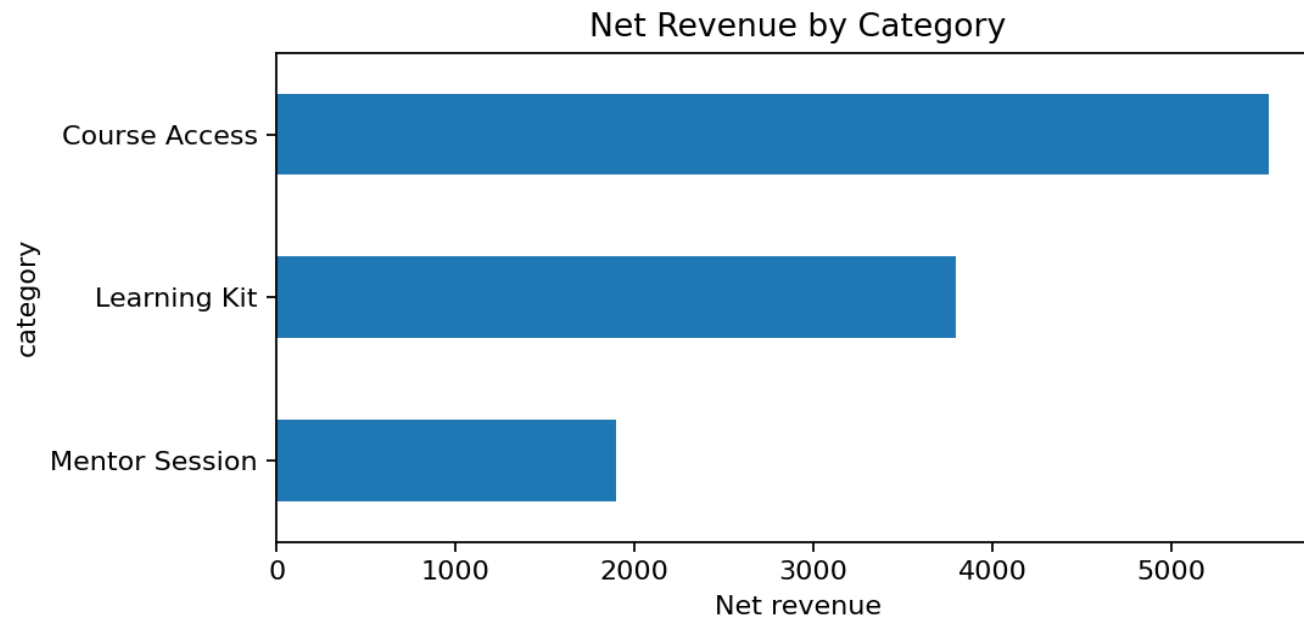
Standardized fields using the supplied data dictionary; missing discount treated as 0%; duplicate order IDs removed before analysis.

EDA: descriptive statistics, distributions, IQR outliers, correlation analysis, and multivariate segment/category/payment views.

Hypotheses: Mann–Whitney U, Kruskal–Wallis and Fisher exact tests at $\alpha = 0.05$.

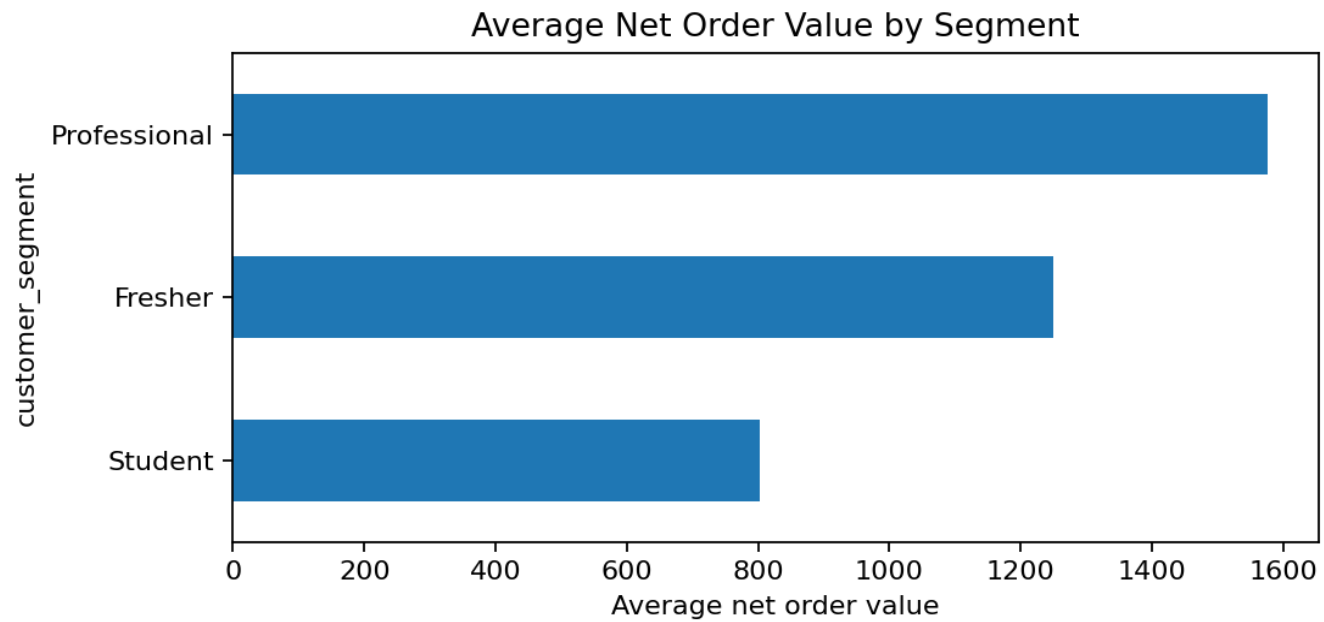
Governance: sample size is too small for high-confidence population inference.

3. Revenue drivers



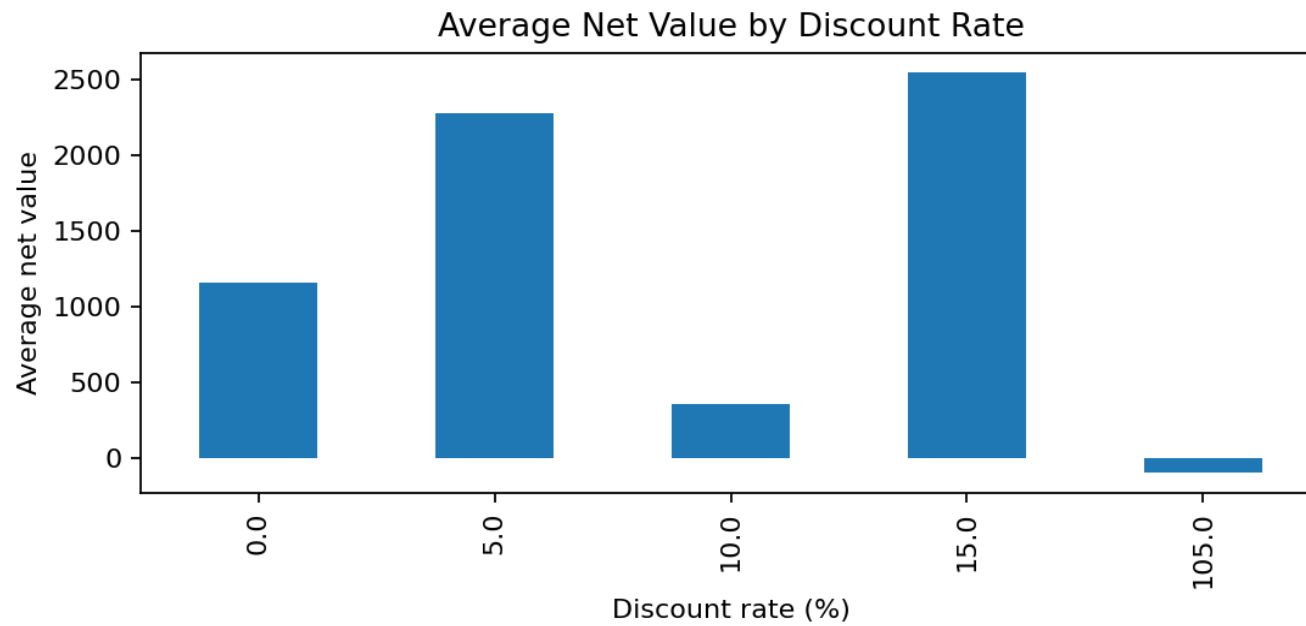
Prioritize Course Access while validating whether concentration persists in a larger time window.

4. Customer-segment opportunity



Professional currently leads average net order value; use as a targeting signal, not a permanent ranking.

5. Discount economics



Discounts lower realized revenue by construction. Incremental demand and margin must be measured before scaling discount policy.

6. Risk areas

- Small-sample risk: 11 unique orders.
- Category concentration can increase product-mix dependence.
- Blanket discounts can create revenue leakage.
- Pending/failed/refunded orders create conversion and cash-realization risk.
- No customer ID exists, so retention cannot be directly measured.

7. Hypothesis test scorecard

H1 — Discount vs net value: $U = \text{nan}$, $p = \text{nan}$ → not significant at 5%.

H2 — Segment vs net value: $H = \text{nan}$, $p = \text{nan}$ → not significant at 5%.

H3 — Discount presence vs Paid: Fisher $p = 0.545$ → not significant at 5%.

Treat results as directional signals; larger samples are required before material pricing/retention changes.

8. Recommendation #1 — Precision discounting

Decision: replace broad discounts with targeted offers by segment/category.

90-day controlled pilot: treatment vs control; monitor conversion, net order value, contribution margin and campaign cost.

ROI: $(\text{incremental contribution} - \text{campaign cost}) / \text{campaign cost}$.

9. Recommendation #2 — Payment conversion

Decision: add payment reminders, alternative payment options and exception monitoring.

Current paid rate: 63.6%. Initial target: +5 percentage points, validated against intervention cost.

ROI: recovered orders × average contribution per order – intervention cost.

10. Recommendation #3 — Retention intelligence

Decision: add customer-level IDs and repeat-purchase tracking.

Track cohort retention, repeat-purchase rate, customer lifetime value, discount dependency and margin.

This creates a repeatable acquisition, retention and pricing decision system.

11. ROI scenarios

Campaign cost, gross margin and reliable retention history are absent; ROI is therefore scenario-based, not forecasted.

5% improvement in realized revenue $\approx$ ■562 incremental revenue before costs/margin.

10% improvement $\approx$ ■1,124 incremental revenue before costs/margin.

Replace scenarios with measured incremental contribution after controlled pilots.

12. Implementation roadmap

0–30 days: data-quality controls, KPI definitions, payment dashboard, customer-ID requirements.

31–60 days: controlled discount experiment and segment/category targeting.

61–90 days: evaluate winners, scale, implement payment recovery and retention cohorts.

90+ days: automate BI monitoring and refresh tests on substantially larger data.

13. C-suite decision checklist

1. Approve a controlled precision-discounting pilot.
2. Fund payment-recovery workflow and weekly paid-rate monitoring.
3. Make customer-level retention measurement a data-model priority.
4. Require margin, campaign-cost and incremental-lift evidence before scaling pricing changes.

14. Capstone portfolio

- Executive Decision Report PDF — this document
- Final Analytical Jupyter Notebook — reproducible EDA, charts, tests and markdown commentary
- Source CSVs — retail orders and data dictionary
- Dashboard link — not supplied; notebook is the reproducible analytical artifact for mentor evaluation.